

In the world of formal verification, **Timed Automata (TA)** extend traditional finite state machines by adding real-valued clocks. Because clocks can take an infinite number of values, we use **Clock Regions** and **CTL (Computation Tree Logic)** to make the infinite space finite and searchable. 

1. Clock Regions: Taming Infinity

The state space of a timed automaton is technically infinite because clocks (e.g., x and y) are continuous. However, the behavior of the system only changes when a clock hits an integer threshold defined in the guards (like $x \leq 5$). 

Clock Regions are an equivalence relation that partitions the infinite state space into a **finite** number of sets. Two clock valuations are in the same region if: 

- Their integer parts are the same.
- The ordering of their fractional parts is the same (this determines which clock will hit the next integer first).

Why Regions Matter

By grouping infinite points into these "bubbles," we create a **Region Automaton**. This is a finite transition system that is "bisimilar" to the timed automaton, allowing us to use standard model-checking algorithms.

2. CTL Properties in Timed Automata

CTL (Computation Tree Logic) is used to describe the branching nature of time. In the context of Timed Automata, we use it to ask if certain timing requirements are met. 

The Building Blocks

CTL uses two-part operators: a **Path Quantifier** followed by a **Temporal Operator**.

Quantifier	Meaning	Temporal Operator	Meaning
A	On A ll paths	G	G lobally (Always)
E	There E xists a path	F	In the F uture (Eventually)
		U	U ntil
		X	Ne X t state



Esporta in Fogli



Common CTL Patterns for Timed Systems

1. **Safety** ($AG \neg \text{bad}$): "On all possible executions, we never enter a 'bad' state (like a crash)."
2. **Liveness** ($AF \text{goal}$): "On all paths, we will eventually reach the 'goal' state."

3. **Reachability** (EF_{success}): "Is there at least one sequence of timings that leads to 'success'?"
 4. **Invariance** ($A[\Box \phi]$): "Property ϕ holds for every single reachable state."
-

3. The Reality Check: TCTL

Standard CTL doesn't allow you to put explicit time bounds inside the formula (e.g., "Will it happen within 5 seconds?"). For that, we use **TCTL (Timed CTL)**.

Instead of just saying AFp , we say:

$$AF_{\leq 5} p$$

This translates to: "On all paths, p will eventually be true **within 5 time units**."